

CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:143.175.143	

TABLE 6

Model (A3) > Geometry > bluff_body_cylinder_noBlockage_OperatingConditions_2 > Parts

Object Name	WakeZone	FluidDomain	FluidDomain	Outer_WakeZone
State	Meshed	Suppressed	Meshed	
Graphics Properties				
Visible	Yes	No	Yes	
Transparency	1			
Definition				
Suppressed	No	Yes	No	
Dimension	2D			
Coordinate System	Default Coordinate System			
2D Behavior	Plane Stress			
Thickness	0. m			
Thickness Mode	Refresh on Update			
Treatment	None			
Material				
Assignment				
Fluid/Solid	Fluid			
Bounding Box				
Length X	2.5 m	1.e-001 m	6. m	4.148 m
Length Y	1. m	0.1 m	3.5 m	1.802 m
Properties				
Volume	0. m³			
Centroid X	-0.24765 m	-1. m	0.13816 m	-0.25 m
Centroid Y	-7.5915e-018 m	-7.1434e-017 m	6.9772e-017 m	-2.2317e-017 m
Centroid Z	0. m			
Surface Area(approx.)	2.4921 m²	7.854e-003 m²	13.525 m²	4.9747 m²
Statistics				
Nodes	113246	0	13694	83226
Elements	217770	0	25720	163778
Mesh Metric	Orthogonal Quality			
Min	0.59213	0.	0.64156	0.64641
Max	1.	0.	1.	
Average	0.97009	0.	0.9653	0.97054
Standard Deviation	3.2293e-002	0.	3.3746e-002	3.0679e-002

TABLE 7

Model (A3) > Materials

Object Name	Materials
State	Fully Defined
Statistics	
Materials	0
Material Assignments	0

Coordinate Systems

TABLE 8

Model (A3) > Coordinate Systems > Coordinate System

Object Name	Global Coordinate System	Coordinate System
State	Fully Defined	
Definition		
Type	Cartesian	
Coordinate System ID	0.	Program Controlled
Coordinate System		
APDL Name		
Suppressed		No
Origin		
Origin X	0. m	-1. m
Origin Y	0. m	
Origin Z	0. m	
Define By		Named Selection
Named Selection		Cylinder
Directional Vectors		
X Axis Data	[1. 0. 0.]	
Y Axis Data	[0. 1. 0.]	
Z Axis Data	[0. 0. 1.]	
Transfer Properties		
Source		
Read Only	No	
Principal Axis		
Axis		X
Define By		Global X Axis

Orientation About Principal Axis		
Axis		Y
Define By		Default
Transformations		
Base Configuration		Absolute
Transformed Configuration		[-1. 0. 0.]

Connections

TABLE 9
Model (A3) > Connections

Object Name	Connections
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	0
Active Contacts	0
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

Mesh

TABLE 10
Model (A3) > Mesh

Object Name	Mesh
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	CFD
Solver Preference	Fluent
Element Size	5.e-002 m
Export Format	Standard
Export Preview Surface Mesh	No
Sizing	
Use Adaptive Sizing	No
Growth Rate	1.15
Mesh Defeaturing	Yes
Defeature Size	1.e-003 m
Capture Curvature	No
Capture Proximity	No
Enable Washers	No
Bounding Box Diagonal	6.9462 m
Average Surface Area	5.25 m ²
Minimum Edge Length	0.31416 m
Quality	
Check Mesh Quality	Yes, Errors
Target Skewness	Default (0.9)
Smoothing	Medium
Mesh Metric	Orthogonal Quality
Min	0.59213
Max	1.
Average	0.96997
Standard Deviation	3.1773e-002
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	2
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	
Rigid Body Behavior	Dimensionally Reduced

Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Use Sheet Thickness for Pinch	No
Pinch Tolerance	Default (4.5e-004 m)
Generate Pinch on Refresh	No
Sheet Loop Removal	No
Automatic Methods	
Sheet Body Method	Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	207492
Elements	407268
Show Detailed Statistics	No

TABLE 11
Model (A3) > Mesh > Mesh Controls

Object Name	cylinder_edge_sizing	Inflation	Inner_wake_sizing	outer_wake_sizing	FD_sizing	sphere of influence	Inflation 2	Inflation 3	Edge Sizing	Edge Sizing 2	Triangles Method
State	Fully Defined				Suppressed	Fully Defined	Suppressed			Fully Defined	
Scope											
Scoping Method	Named Selection	Geometry Selection									
Named Selection	Cylinder										
Geometry		1 Face							4 Edges	2 Edges	3 Bodies
Definition											
Suppressed	No				Yes	No	Yes			No	
Type	Element Size		Element Size			Sphere of Influence		Element Size	Number of Divisions		
Element Size	1.2e-003 m		5.e-003 m	8.e-003 m	5.e-002 m	3.e-003 m		5.e-003 m			
Boundary Scoping Method		Named Selections					Geometry Selection				
Boundary		Cylinder					4 Edges				
Inflation Option		First Layer Thickness					First Layer Thickness				
First Layer Height		1.8e-005 m					5.e-003 m	9.e-003 m			
Maximum Layers		27					3	10			
Growth Rate		1.2					1.2				
Inflation Algorithm		Pre					Pre				
Active					No, Suppressed		No, Suppressed				
Sphere Center						Coordinate System					
Sphere Radius						7.e-002 m					
Control Messages						No					
Number of Divisions								100			
Method											Triangles
Element Order											Use Global Setting
Advanced											
Behavior	Hard		Soft						Soft		
Capture Curvature	No		No						No		
Capture Proximity	No		No						No		
Bias Type	No Bias							No Bias			
Blend to Neighbors	Yes							Yes			
Defeature Size			1.e-003 m	Default (1.e-003 m)							
Growth Rate			Default (1.15)						Default (1.15)		
Bias Option									Bias Factor		
Bias Factor									Please Define		
Reverse Bias									No Selection		

Advanced Improve Options		
Aggressive Thin Face Collapse		Program Controlled
Automatic Node Movement		Program Controlled
Refinement Options		
Refine at Thin Section		No

Named Selections

TABLE 12
Model (A3) > Named Selections > Named Selections

Model (A3) > Named Selections > Named Selections					
Object Name	Inlet	Outlet	Upper_Wall	Lower_Wall	Cylinder
State	Fully Defined				
Scope					
Scoping Method	Geometry Selection				
Geometry	1 Edge				
Definition					
Send to Solver	Yes				
Protected	Program Controlled				
Visible	Yes				
Program Controlled Inflation	Exclude				
Statistics					
Type	Manual				
Total Selection	1 Edge				
Length	3.5 m		6. m		0.31416 m
Suppressed	0				
Used by Mesh Worksheet	No				